# Dataset Declaration

**Project Title:** PakRoad FloodWatch (Climate & Infrastructure Analytics) | **Track:** Climate (Rainfall & Urban Road Inundation Risk)

**Team Name:** [Your Team Name] | **Submission Deadline:** Monday, 19 October 2026, 12:00 PM PKT

## Source 1: Primary Real-Time & Historical Weather & Hydrological APIs

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| Name & Publisher: | Open-Meteo Historical Weather & GloFAS Flood APIs • *Open-Meteo GmbH & Copernicus EMS* |
| URL & Licence: | https://open-meteo.com/en/docs/historical-weather-api \| https://flood-api.open-meteo.com • *CC BY 4.0* |
| Update Frequency: | Hourly real-time updates + continuous historical archive (2021–2026) |
| Variables Extracted: | `temperature_2m` (°C), `relative_humidity_2m` (%), `dew_point_2m` (°C), `precipitation` (mm), `rain` (mm), `surface_pressure` (hPa), `soil_moisture_0_to_7cm` ($m^3/m^3$), `soil_moisture_7_to_28cm` ($m^3/m^3$), `wind_speed_10m` (km/h), `wind_direction_10m` (°), `cloud_cover` (%), `weather_code` (WMO), and `river_discharge` ($m^3/s$). |

## Source 2: OpenStreetMap Spatial Infrastructure & Road Connectivity

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| Name & Publisher: | OpenStreetMap / Geofabrik GIS & HeiGIT HDX Risk Assessment Indicators • *OSMF, HOT OSM & HeiGIT / UN OCHA* |
| URL & Licence: | https://download.geofabrik.de/asia/pakistan.html \| https://data.humdata.org/dataset/pakistan---risk-assessment-indicators • *ODbL* |
| Update Frequency: | Continuous community mapping & periodic official humanitarian releases |
| Attributes Extracted: | Nationwide road networks, emergency healthcare facilities & hospitals (GeoJSON), populated places & settlements, district Rural Access Index (RAI - % population with all-season road connectivity within 2km), multi-tier flood return period exposure (RP10, RP50, RP100), hospital travel access times, and evacuation buffer capacity across all 160 Pakistani districts. |

## Source 3: Government Hazard Advisories, Bulletins & Early Warning Triggers

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| Name & Publisher: | NDMA Pakistan & PMD Flood Forecasting Division • *Government of Pakistan* |
| URL & Licence: | https://ndma.gov.pk \| https://ffd.pmd.gov.pk • *Official Public Domain / Open Government Release* |
| Update Frequency: | Daily flood bulletins, real-time weather advisories & multi-year seasonal strategies |
| Attributes Extracted: | Official monsoon advisories (Aug–Oct 2026), Pakistan National Anticipatory Action Strategy (Early Warning & Trigger Systems 2026–2030), daily river discharge bulletins, and river flood stage danger limits (Low, Medium, High, Very High) across 19 critical stations on Indus, Jhelum, Chenab, Ravi, and Kabul river basins. |

## Source 4: National Demographic, Agricultural & Socioeconomic Baselines

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| Name & Publisher: | Pakistan Bureau of Statistics (PBS) Official Census & Economic Indicators • *Ministry of Planning Development* |
| URL & Licence: | https://www.pbs.gov.pk • *Government Open Statistical Release* |
| Update Frequency: | Decennial digital census releases, weekly SPI price indices & quarterly national accounts |
| Attributes Extracted: | 7th Digital Population & Housing Census 2023 district summaries (Punjab & Sindh flashcards), 160-district population demographics, urban/rural distribution, vulnerability & dependency ratios, Integrated Agricultural Census (IAC) country report, and Consumer Price Index (CPI / SPI) food/energy price indicators. |

## Which sources are not on the track list, and why are they credible?

> **Sources 2, 3, and 4 (OpenStreetMap/HDX, NDMA/PMD, and PBS)** are not listed on the default Climate track benchmark list, but provide indispensable, authoritative empirical ground truth:

- **OpenStreetMap / Geofabrik & HDX (HeiGIT):** Hosted on UN OCHA's Humanitarian Data Exchange platform and curated by Heidelberg University, serving as the international gold standard for spatial infrastructure and emergency road accessibility.
- **NDMA & PMD Flood Forecasting Division:** The constitutional federal agencies governing emergency response and hydrometeorology in Pakistan, providing legal threshold definitions for river flood stages.
- **Pakistan Bureau of Statistics (PBS):** The official national statistical authority of Pakistan, providing certified Digital Census 2023 demographics and economic indices.

## Expected Volume

- **Meteorological Heat & Rainfall Data:** Continuous hourly observations across **12 major Pakistani metropolitan areas and economic hubs** (Lahore, Karachi, Rawalpindi, Islamabad, Faisalabad, Multan, Peshawar, Quetta, Hyderabad, Sukkur, Gilgit, Gwadar) spanning nearly 6 full years (2021–2026), capturing the historic 2022 super floods:

  > **Total Ingested Meteorological Volume:** **592,176 hourly records** *(Exceeds 50,000-row requirement by >11x)*

- **Hydrological Streamflow Data: 10,585 daily river discharge readings** across 5 critical river basins (Ravi, Indus, Kabul, Soan, Malir).
- **Spatial & Road Infrastructure:** Full nationwide coverage across **160 administrative districts** (Rural Access Index, road inundation exposure, healthcare travel buffers, and GeoJSON hospital/settlement footprints).
- **Official Hazard Reports & Census Baselines:** Multi-year NDMA anticipatory action triggers, PMD river danger limits, and 2023 Digital Census district demographic datasets.

## Why this data fits the problem and the primary user

- **Primary Users:** City Traffic Police departments (Lahore, Karachi, Rawalpindi) and Rescue 1122 emergency dispatchers.
- **Operational Fit:**
  1. **Meteorological Saturation (Open-Meteo):** Real-time and forecasted precipitation, rain intensity, and antecedent soil moisture determine surface runoff volume and localized ponding potential.
  2. **Hydrological Backflow (GloFAS & PMD):** Identifies when main rivers (Ravi, Indus, Malir, Soan) exceed high-flood thresholds, choking municipal storm outfalls and reversing drainage flow.
  3. **Spatial Vulnerability (OpenStreetMap & HDX):** Pinpoints road links lacking all-season paving (low RAI) and routes cut off from emergency healthcare facilities.
  4. **Demographic & Trigger Context (NDMA & PBS):** Maps population density and institutional early warning triggers to deploy de-watering pumps and enforce traffic diversions 2 to 4 hours before severe road submergence.

## Baseline you declare (Fixed before results)

- **Precipitation Baseline:** 24-hour persistence model:

  > $\hat{y}_t = y_{t-24h}$
  >
  > (The predicted precipitation at hour *t* equals the observed precipitation at the same hour on the previous day).

- **Road Condition Baseline:** Static heuristic rule:

  > • **Waterlogging Risk / Caution:** Triggered if hourly rainfall ≥ 2.5 mm/hr | • **Normal / Passable:** Triggered when hourly rainfall < 2.5 mm/hr

- **Success Criterion:** The trained predictive machine learning models (LightGBM / XGBoost Regressor and Classifier) must achieve at least a **20% improvement** in Mean Absolute Error (MAE) and Brier Score over the declared persistence baseline on an untouched future temporal evaluation set (2025–2026).